

The Brazilian Agenda of Economic Development

Macroeconomic stability, productivity, state capacity, and inclusion

Eduardo Peres Cunha | PUC-Rio – Brazilian Seminars | July 2026

The lecture starts from one puzzle

Why has a large, urban, resource-rich economy with strong institutions in selected areas failed to achieve sustained income convergence?

Past achievement

Rapid industrialization and urbanization.

Institutional achievement

Stabilization, social policy, agriculture, and digital payments.

Unfinished transition

Low aggregate productivity and persistent inequality.

Brazil combines frontier capabilities with large deficits

Continental scale

More than 200 million people and 5,500 municipalities.

Institutional innovation

Pix scaled digital public infrastructure to most adults.

Clean electricity

86.6% of domestic electricity supply was renewable in 2025.

Persistent duality

38.1% of workers remained informal in 2025.

Source: World Bank, Brazil Overview; EPE, BEN 2026; BCB, Pix statistics; IBGE, PNAD Continua 2025.

Development is broader than current GDP growth

Income	Productivity	Inclusion	Resilience
Income per person	Output per worker	Mobility and access	Climate and institutions

$$\frac{Y}{N} = \underbrace{\frac{Y}{L}}_{\text{output per worker}} \times \underbrace{\frac{L}{N}}_{\text{workers per person}}$$

- ▶ Productivity determines how much each worker produces.
- ▶ Demography and labor-force participation determine how many people work.
- ▶ Distribution determines who can invest in skills and participate.

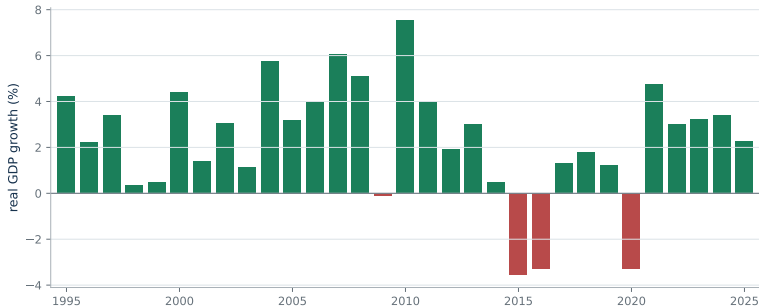
Roadmap for 150 minutes

Time	Block	Main question
0–12 min	The puzzle	What counts as development?
12–65 min	Macro foundations	How do inflation, fiscal policy, and exchange rates shape growth?
65–75 min	Exchange-rate case	What changed in 1999, and why does the regime matter?
75–85 min	Break	
85–135 min	Micro constraints	Firms, workers, infrastructure, security, and state capacity
135–147 min	Integrated agenda	Which reforms are complements?
147–150 min	Questions	Which constraint changes the rest of the agenda?

Part I – Making growth possible

Macroeconomic stability, exchange-rate regimes, fiscal space, and investment

Growth has been volatile and convergence has been intermittent



- ▶ Brazil combined strong expansions with three recessions after 1995.
- ▶ The post-2010 problem is not one crisis alone: average growth weakened before the pandemic.

Source: BCB SGS 7326, based on IBGE National Accounts. Annual real GDP growth; 2025 is the latest observation.

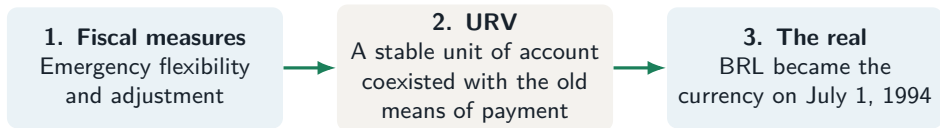
High inflation disrupts more than purchasing power



- ▶ Stabilization restored price information and lengthened planning horizons.
- ▶ The later spikes show why credibility, fiscal policy, and the exchange rate still interact.

Source: BCB SGS 433, monthly IPCA. Twelve-month rates are compounded from the official monthly series.

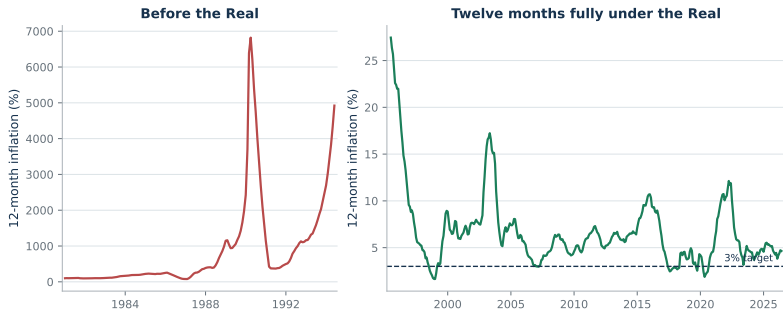
The Real Plan changed the unit of account before the currency



- ▶ The URV coordinated prices without an overnight freeze.
- ▶ Its value was linked to the U.S. dollar during the transition.
- ▶ The exchange rate then became part of the nominal anchor.

Source: BCB, [The Real Plan](#).

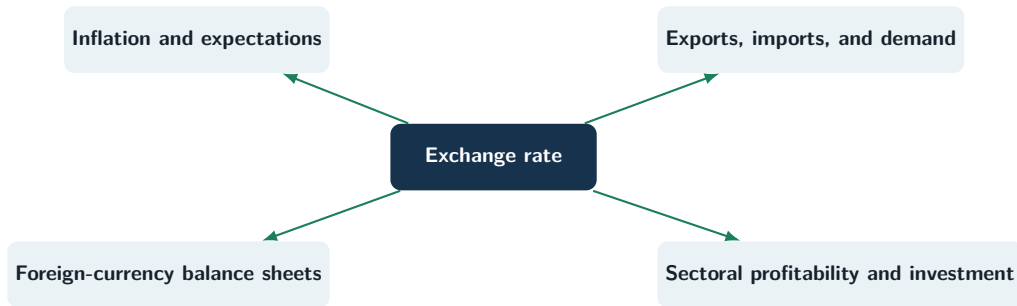
The Real Plan produced a discrete break in inflation



- ▶ Twelve-month inflation reached thousands of percent before July 1994.
- ▶ From July 1995 onward, every month in the twelve-month window uses prices quoted under the Real.

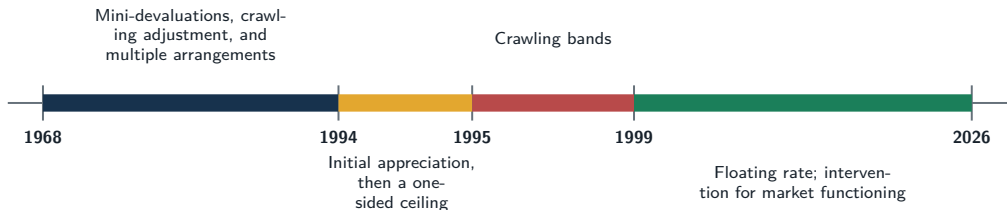
Source: BCB SGS 433, monthly IPCA. Rates are compounded over twelve months. The two panels use different vertical scales.

The exchange rate links the macroeconomy to development



An exchange-rate regime allocates adjustment across the currency, interest rates, reserves, and domestic demand.

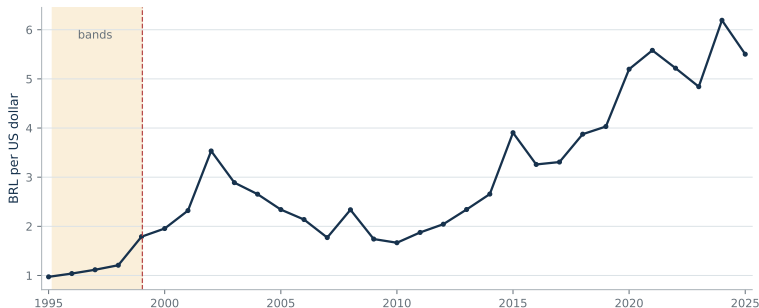
Brazil moved from managed adjustment to a formal float



- ▶ Historical labels summarize changing rules and multiple market segments.
- ▶ The operational regime can differ from a simple fixed-versus-floating label.

Source: BCB, 1999 FX Market Analysis and official FX-policy pages; dates before 1994 are a schematic summary.

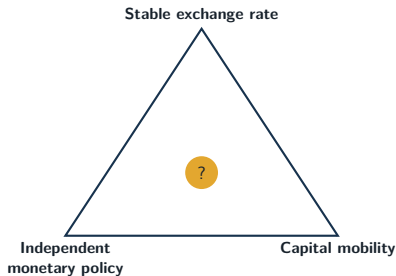
The exchange rate adjusted after the 1999 regime change



- The band limited adjustment before 1999; the float then absorbed domestic and external shocks.
- A weaker real is not a development policy by itself: relative prices, balance sheets, and persistence matter.

Source: BCB SGS 3692. BRL per US dollar, selling rate at year-end; shaded period marks the explicit-band era.

The policy trilemma makes the trade-off explicit



- ▶ A country can fully pursue only two corners at once.
- ▶ Under an exchange-rate anchor, interest rates defend the parity.
- ▶ Under a float, the currency absorbs part of external shocks.
- ▶ Capital-flow measures change the constraint but do not remove it.

The Real used the exchange rate as a nominal anchor

Why it helped

A stable currency price disciplined tradable-goods prices and coordinated expectations after hyperinflation.

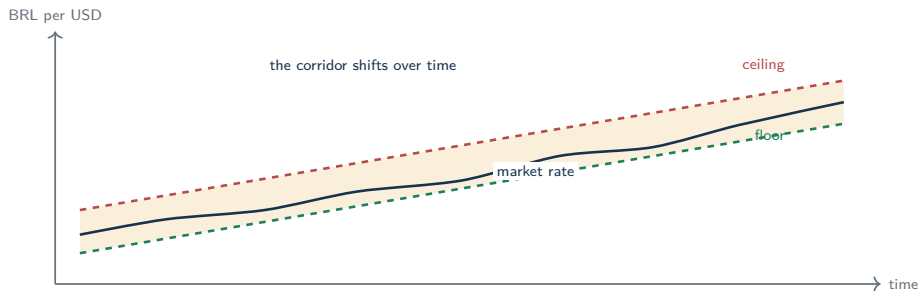
Why pressure accumulated

Domestic inflation exceeded foreign inflation, the real appreciated, and high interest rates attracted capital while defending the regime.

- ▶ July–September 1994 first allowed the real to appreciate below BRL 1 per USD.
- ▶ September introduced a one-sided BRL 1 per USD dollar-sale ceiling, not a fixed parity.
- ▶ Explicit symmetric bands began in March 1995.
- ▶ The currency depreciated gradually inside a managed corridor.
- ▶ External shocks tested reserve and interest-rate defenses.

Source: BCB, FX Market Analysis, 1999 Q1.

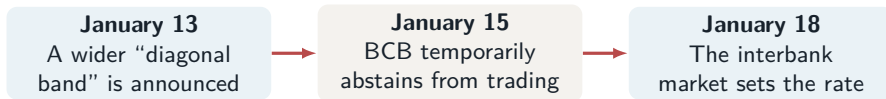
A crawling band trades flexibility for credibility



- The market rate fluctuates inside the corridor while the authorities move its boundaries over time.
- A narrow corridor can strengthen the anchor but invites one-way bets when fundamentals diverge.

Source: Stylized diagram; BCB, 1999 Q1 FX Market Analysis.

January 1999 compressed a regime change into five days



- ▶ The BCB retained the option to intervene against disorderly movements.
- ▶ Formal inflation targeting followed in July 1999.
- ▶ The nominal anchor shifted from the exchange rate to the inflation target.

Source: BCB, [Communiqué 6,565](#); BCB Working Paper 25.

Since 1999, the exchange rate floats and the BCB targets inflation

Formal regime

The market determines the exchange rate.
The BCB does not intervene to select a level.

Monetary anchor

The policy rate responds to the inflation outlook under an explicit target.

Adjustment mechanism

The currency absorbs part of external shocks while interest rates respond to the inflation outlook.

The nominal anchor moved from the exchange rate to the inflation target.

Source: BCB, [Foreign Exchange Policy](#).

Monetary policy reacts to the inflation outlook



- ▶ The policy rate is forward-looking; it need not move one-for-one with current inflation.
- ▶ Fiscal risk and exchange-rate pass-through affect how much tightening is required.

Source: BCB SGS 432 and 433. Selic target is the last observation of each month; IPCA is compounded over 12 months.

Nominal and real exchange rates answer different questions

Nominal rate

$$E = \text{BRL per USD}$$

A rise in E is a nominal depreciation of the real.

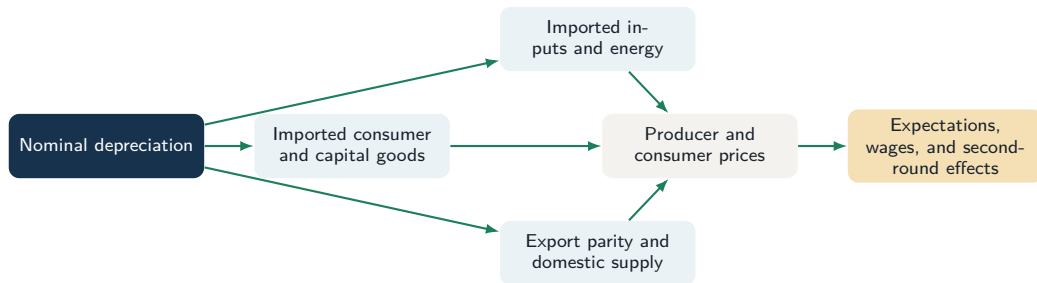
Real exchange rate

$$q = \frac{EP^*}{P}$$

A rise in q makes foreign goods more expensive relative to domestic goods under this definition.

- ▶ Domestic and foreign inflation can undo a nominal depreciation.
- ▶ Productivity, terms of trade, and risk premia affect the equilibrium real rate.
- ▶ “Competitive” has no meaning without a price index, horizon, and sector.

A depreciation reaches inflation through several margins



Pass-through depends on expectations, slack, margins, initial inflation, and the size of the shock.

Source: BCB Working Papers 5, 181, and 473.

Exchange-rate pass-through is state-dependent

Normal conditions

Credible targets, anchored expectations, slack, and smaller shocks limit propagation.

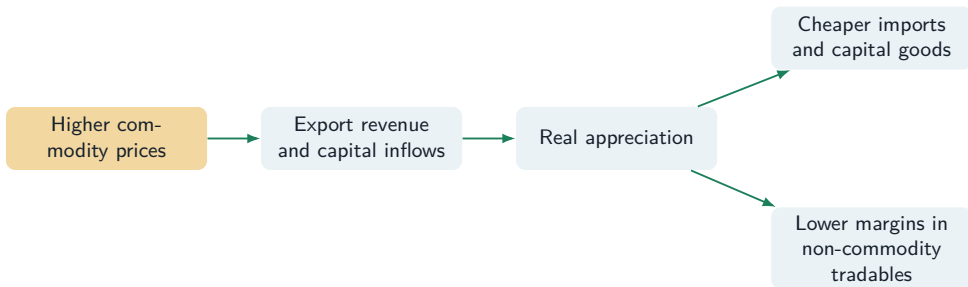
Crisis conditions

Large depreciations, high uncertainty, indexation, and unanchored expectations raise propagation.

- ▶ A 2026 BCB exercise estimates that a 10% depreciation adds about 1.0 percentage point to IPCA over 12 months and 1.6 points over 24 months.
- ▶ The confidence bands are wide, and the effects differ sharply across food, industrial goods, and services.
- ▶ Monetary policy responds to the inflation forecast, not mechanically to the exchange rate.

Source: BCB, March 2026 Monetary Policy Report, exchange-rate pass-through box; estimates are episode-dependent.

Commodities affect the currency and the production structure



- ▶ The exchange rate distributes a terms-of-trade shock across consumers and sectors.
- ▶ “Dutch disease” is a mechanism to test, not a complete diagnosis.

A weaker currency has ambiguous development effects

Margin	Possible gain from depreciation	Possible cost
Exporters	Higher domestic-currency revenue	Imported inputs and financing become costlier
Import-competing firms	More price space	Capital goods and technology cost more
Households	Employment may shift toward tradables	Food, fuel, and durable goods become dearer
Public sector	Some revenues rise with nominal activity	Indexed debt and inflation costs can increase
Productivity	Entry into tradables may support learning	Protection can preserve low-productivity firms

The answer depends on balance sheets, imported-input intensity, credibility, and persistence.

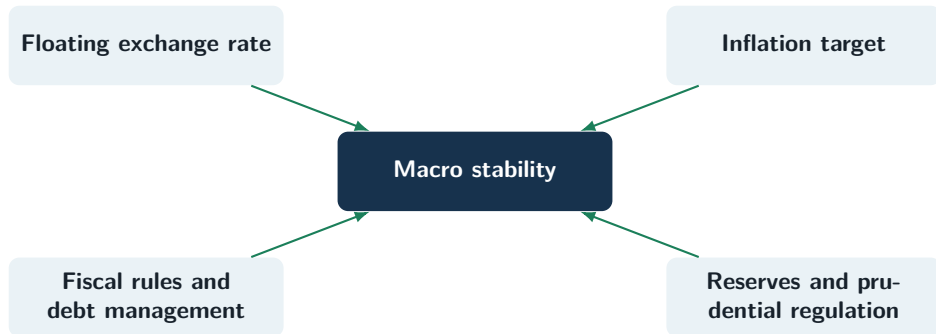
Reserve accumulation changed Brazil's external buffer



- ▶ Reserves insure against external funding stops and reduce foreign-currency liquidity risk.
- ▶ The stock rose sharply before the global financial crisis and has remained large.

Source: BCB SGS 13621. International reserves, cash concept; last daily observation of each month, US\$ billion.

The post-1999 framework combines adjustment and anchors



- ▶ The pieces are complements: fiscal risk affects interest rates and the currency.
- ▶ The float absorbs shocks; monetary credibility limits pass-through.

Brazil's fiscal problem is about composition and credibility

The state is not small

Revenue and social spending are substantial for an upper-middle-income economy.

Much spending is rigid

Pensions, payroll, transfers, and earmarks limit annual reallocation.

Development margin

Public investment and high-return maintenance compete for residual fiscal space.

Macro margin

Uncertain debt paths raise risk premia and shorten private planning horizons.

Source: OECD, [Economic Survey: Brazil 2023](#).

Debt dynamics compress the space for development spending



- ▶ Debt rose before the pandemic, jumped in 2020, and remains high relative to the mid-2000s.
- ▶ Interest rates, primary balances, growth, and debt composition jointly determine the path.

Source: BCB SGS 13762. Gross general government debt, methodology used since 2008, monthly, percent of GDP.

Low saving and investment constrain capital deepening

16.8%
investment / GDP, 2025

14.4%
saving / GDP, 2025

- ▶ Capital per worker grows slowly when investment remains low.
- ▶ High financing costs screen out long-horizon projects.
- ▶ Infrastructure bottlenecks reduce the return to private capital.
- ▶ Productivity growth matters because more investment alone can misallocate resources.

Source: IBGE, National Accounts 2025; rates are shares of GDP.

Macro scorecard: a major achievement and an incomplete transition

Area	Achievement	Remaining constraint
Inflation	Hyperinflation ended; expectations have an explicit anchor	Shocks still pass through the currency and indexation
Exchange rate	A float absorbs external shocks	Volatility affects prices, balance sheets, and investment
External sector	Reserves and lower FX mismatch strengthened buffers	Commodity and global-rate cycles remain important
Fiscal policy	Strong revenue capacity and broad social insurance	Rigid spending and debt uncertainty compress investment
Capital formation	Deep domestic financial markets	High cost of capital and low investment

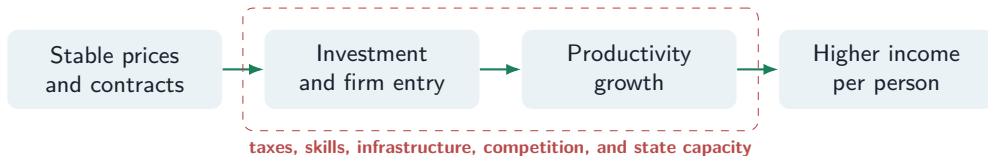
10-minute break

After the break: why stability did not restore convergence

Part II – Making growth happen

Productivity, firms, workers, and the institutions that connect them

Stability raises the return to investment; it does not guarantee it



Brazil solved much of the first arrow. The middle of the chain remains weak.

Aggregate productivity depends on firms and allocation

$$\text{Aggregate productivity} = \underbrace{\text{productivity within firms}}_{\text{technology, management, skills, scale}} + \underbrace{\text{allocation across firms}}_{\text{entry, exit, competition, capital and labor flows}}$$

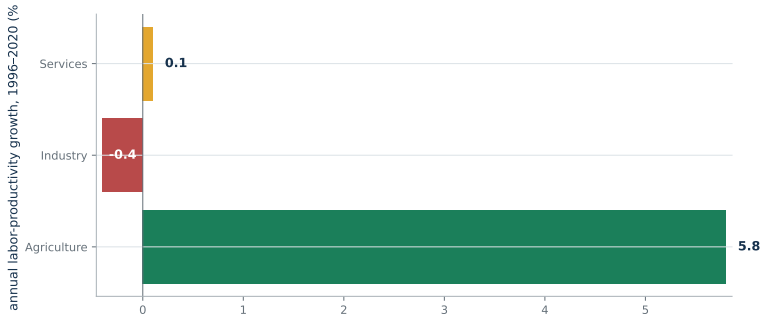
Within-firm margin

Can an existing producer adopt better technology and organization?

Reallocation margin

Do productive firms expand while persistently unproductive firms shrink or exit?

Brazil's productivity problem is not uniform across sectors



- ▶ Agriculture shows that sustained productivity growth is possible.
- ▶ Services employ most workers, so diffusion beyond the frontier sector determines the aggregate result.

Source: World Bank, labor-productivity growth by sector, 1996–2020; values rounded.

A long tail of low-productivity firms shapes aggregate outcomes

The long tail

Many small producers use little capital, management technology, or formal finance.

The frontier

A smaller set of firms competes globally and adopts advanced technology.

- ▶ Informality lowers tax and regulatory costs but limits scale and credit access.
- ▶ Complex rules encourage firms to remain small or split activities.
- ▶ Weak exit and reallocation preserve low-productivity uses of capital and labor.

Firm capabilities determine whether technology is adopted

Management

Targets, inventory control, quality systems, and worker incentives.

Technology

Digital tools pay off only when processes and skills change with them.

Scale

Formal finance and access to larger markets allow fixed costs to be spread.

Credit subsidies alone do little when management, competition, and demand remain unchanged.

Source: World Bank, Brazil productivity diagnostics; OECD, Economic Survey: Brazil 2023.

Informality is both insurance and a productivity trap

Why workers and firms enter

Low entry costs, flexible hours, weak formal-job creation, and immediate income.

What it provides

A margin of adjustment when formal employment and social insurance do not reach everyone.

What it limits

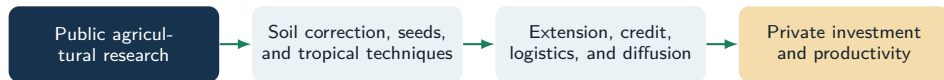
Credit, training, scale, legal enforcement, tax capacity, and portable protection.

Policy margin

Lower formalization cliffs and make benefits portable across employment types.

Source: IBGE, PNAD Continua; OECD, Economic Survey: Brazil 2023. Informality is a statistical labor-market category, not a synonym for tax evasion.

Embrapa shows how complementary institutions raise productivity



Lesson

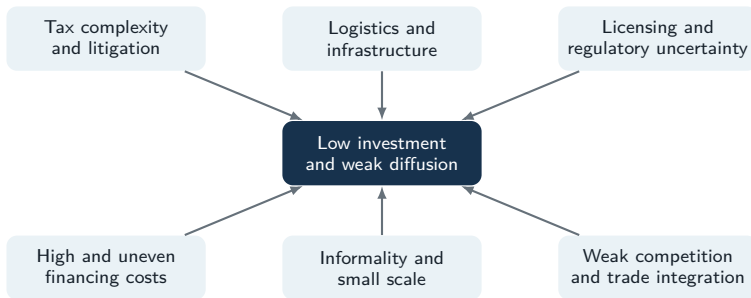
Mission-oriented research works when adoption, finance, and infrastructure reinforce it.

Trade-off

Productivity per hectare and environmental enforcement jointly determine land-use outcomes.

Source: Embrapa Cerrados; Ipea, agricultural productivity.

“Custo Brasil” is a system of interacting wedges



Removing one wedge may have little effect when the other complements remain binding.

The consumption-tax reform is a live implementation test

2026	2027	2029–32	2033
Test year for CBS and IBS.	CBS begins; PIS/Cofins end.	Gradual ICMS/ISS transition.	Full new system.

- ▶ A destination-based VAT should reduce cascading taxes and production distortions.
- ▶ Federal coordination, exemptions, compliance systems, and transition rules determine the gain.
- ▶ Implementation is an economic reform, not an administrative footnote.

Source: Receita Federal, [Consumption Tax Reform](#).

Competition policy changes the return to innovation

When entry is credible

Incumbents invest, lower markups, and adopt technology to defend market share.

When rents are protected

Licensing, procurement design, network bottlenecks, and local political connections can preserve inefficient firms.

- ▶ Antitrust, bankruptcy, trade exposure, and transparent procurement affect the same reallocation margin.
- ▶ The objective is contestability, not a mechanical preference for large or small firms.

Source: OECD, Competition Assessment Reviews: Brazil; CADE, institutional reports.

Logistics determine which firms can reach national markets

Distance

A continental economy makes transport time and reliability part of marginal cost.

Networks

Roads, rail, ports, storage, and digital systems are complements.

Governance

Project selection, maintenance, concessions, and licensing determine returns.

A new asset raises productivity only if the network around it works.

Source: World Bank, Brazil infrastructure diagnostics; EPL/Infra S.A., National Logistics Plan.

Sanitation is infrastructure and human-capital policy

Household channel

Water quality and sewage coverage affect disease, school attendance, time use, and property values.

Implementation channel

Municipal scale, regulation, contracts, and service affordability shape expansion.

- ▶ National averages conceal large regional and within-city gaps.
- ▶ Universalization requires investment and credible regulation, not construction alone.

Source: SNIS, Ministry of Cities; IBGE, 2022 Population Census sanitation results.

Trade can raise productivity through inputs, scale, and selection

Channels

- ▶ Cheaper and better intermediate inputs
- ▶ Larger markets for productive exporters
- ▶ Technology and management diffusion
- ▶ Entry and exit under competition

Transition costs

- ▶ Workers and places adjust slowly
- ▶ Logistics and customs can bind first
- ▶ A volatile real exchange rate changes incentives
- ▶ Skills determine who captures the gains

The relevant choice is the sequence and support for adjustment, not autarky versus free trade.

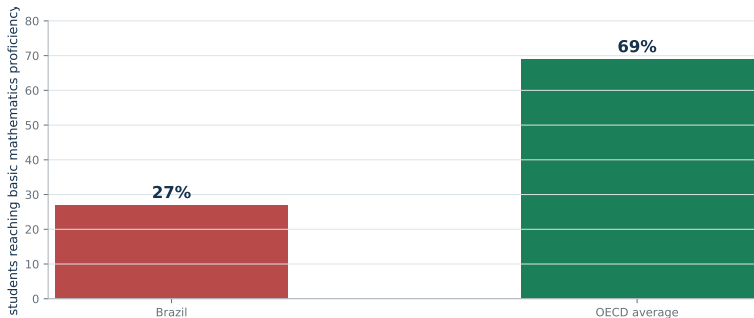
The credit spread remains a microeconomic wedge



- ▶ The spread combines expected losses, operating costs, taxes, market power, and risk premia.
- ▶ Better collateral, information sharing, competition, and insolvency rules address different components.

Source: BCB SGS 20783. Average spread on outstanding credit, monthly, percentage points.

The human-capital constraint is learning, not enrollment alone



- ▶ Learning gaps reduce adoption of technology and worker mobility.
- ▶ Teacher practice, early childhood, completion, and school governance operate at different horizons.

Source: OECD, PISA 2022: Brazil.

Human capital starts before school and differs by place

Early childhood

Nutrition, health, language exposure, and childcare shape readiness.

School system

Teacher support and instructional time determine whether enrollment becomes learning.

Territory

Transport, violence, sanitation, and local fiscal capacity affect access and continuity.

Equal spending does not imply equal learning when local constraints differ.

Source: World Bank, Brazil Human Capital Review; IBGE and Inep education indicators.

Low unemployment can coexist with a dual labor market



Source: BCB SGS 24369, sourced from IBGE PNAD Continua; IBGE annual informality rate for 2025.

Childcare expands labor supply and children's capabilities

Labor-market effect

Affordable, reliable childcare raises mothers' ability to take formal jobs and invest in careers.

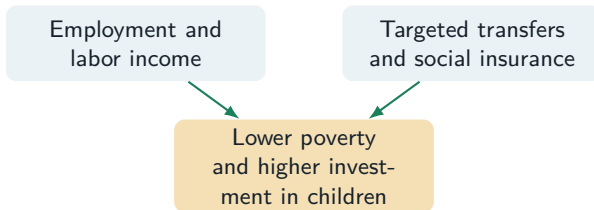
Human-capital effect

Quality matters: childcare can improve language, nutrition, and school readiness.

- ▶ Coverage, hours, location, and quality jointly determine take-up.
- ▶ The return is highest when expansion targets families with the largest access gaps.

Source: IBGE, PNAD Continua care responsibilities; World Bank, Brazil gender and human-capital diagnostics.

Growth and redistribution are not opposing mechanisms



- ▶ Recent poverty reduction reflects both labor income and transfers.
- ▶ Benefit design should preserve coverage while reducing abrupt formalization penalties.
- ▶ Poverty rates must always report the line and purchasing-power standard used.

Source: Ipea, Poverty and Inequality, 1995–2024; IBGE, Social Indicators 2025.

Pix is a case of digital public infrastructure

170m+
individual users

7bn+
transactions in January 2026

- ▶ The central bank set common rails, rules, and access standards.
- ▶ Private institutions compete on top of a public protocol.
- ▶ Immediate settlement lowers payment frictions for households and small firms.
- ▶ Fraud, digital access, and over-indebtedness remain policy concerns.

Source: BCB, [Pix Statistics](#); figures reported in early 2026.

Spatial inequality changes which reform binds first

Productive structure

Agriculture, industry, services, and public employment have different regional weights.

Local capacity

Municipalities differ in tax bases, technical staff, procurement, and service networks.

Household access

Transport, schools, health, sanitation, and security shape mobility within the same city.

A national reform produces local results through unequal implementation capacity.

Source: IBGE, Regional Accounts and 2022 Population Census; Ipea, regional development studies.

State capacity is an economic input

Administrative capacity

Select projects, hire staff, procure inputs, manage contracts, and use data.

Legal capacity

Enforce contracts, protect rights, resolve disputes, and sanction corruption.

Fiscal capacity

Raise revenue with low compliance costs and allocate it toward high-return services.

Coordination capacity

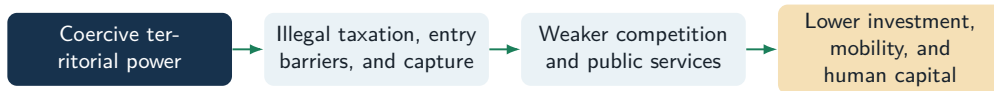
Align federal, state, and municipal responsibilities across shared networks.

Source: World Bank, The Brazil of the Future; OECD, Government at a Glance: Latin America and the Caribbean.

Micro lens – Security and development

How coercive territorial power changes markets, services, and state capacity

Organized crime is a microeconomic constraint



- ▶ Extortion acts like an illegal tax; forced suppliers create local market power.
- ▶ Land appropriation weakens property rights and collateral.
- ▶ Intimidation and corruption can redirect enforcement and public resources.

Source: MPRJ documented cases; SENAPPEN; GENI/UFF and Instituto Fogo Cruzado. Mechanisms are not estimates of aggregate GDP losses.

There is no single organized-crime statistic

Lethal violence

Unit: deaths

National and annual. It does not identify the organization responsible.

Prison intelligence

Unit: groups

SENAPPEN identified at least 88 organizations affecting prisons in its 2024 exercise.

Territorial control

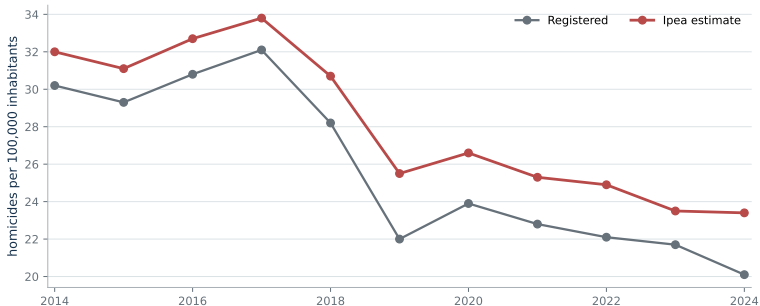
Unit: places and residents

Directly measures armed governance, but only for Metropolitan Rio.

Measurement must match the mechanism being discussed.

Source: Ipea/FBSP, Atlas da Violencia 2026; SENAPPEN, Mapa das ORCRIMs; GENI/UFF and Instituto Fogo Cruzado, 2025.

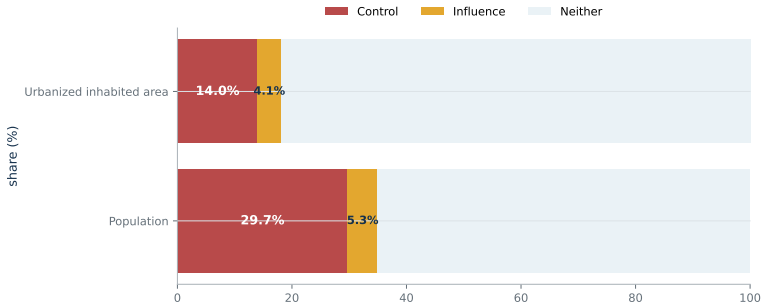
Lethal violence fell, but data quality changes the trend



- ▶ The registered rate fell after 2017; reallocating some deaths of undetermined cause produces a smaller decline.
- ▶ Homicide measures lethal violence. It does not measure the size or revenue of criminal organizations.

Source: Ipea and FBSP, Atlas da Violência 2026, based on SIM/Ministry of Health and IBGE. Rates per 100,000. Estimated series is model-based.

Metropolitan Rio shows the scale of territorial armed governance



- In 2024, 34.9% of residents were estimated to live under armed-group control or influence.
- The estimate covers Metropolitan Rio, uses anonymous reporting, and is not a national census.

Source: GENI/UFF and Instituto Fogo Cruzado, Mapa Historico dos Grupos Armados 2025, with 2022 Census population data. Published shares are rounded.

Coercive market power reaches ordinary goods and services

Household services

Cooking gas, broadband, cable TV, and electricity.

Mobility

Alternative transport and compulsory payments from drivers.

Property

Irregular construction, forced dispossession, and property sales.

Local retail

Security fees, merchant extortion, and forced suppliers.

Economic wedges

- ▶ illegal tax
- ▶ restricted entry
- ▶ weak property rights
- ▶ distorted procurement

Source: MPRJ cases in Nova Iguacu (2019), Itaboraí (2019), Queimados (2024), and the illegal-construction task force (2022). These cases do not measure national prevalence.

Territorial violence erodes human capital

Exposure

51%

of students in the 19 municipalities studied were affected by armed violence in 2022.

Learning

10 points

maximum reported SAEB gap, interpreted by the authors as about six months of learning.

Dropout

**7.2 vs.
9.0–9.6%**

non-controlled versus controlled areas in Metropolitan Rio.

Source: UNICEF, Educacao Sob Cerco 2025, with GENI/UFF and Instituto Fogo Cruzado. Conditional estimates with statistical controls; the design is not experimental.

A credible response combines enforcement and service delivery

Margin	Instruments	Development outcome
Territorial prevention	Safe routes, reliable transport, service continuity, youth prevention	Attendance, business entry, service uptime
Investigation and justice	Forensics, witness protection, case management, corruption control	Clearance with due process, time to disposition
Financial disruption	Beneficial ownership, intelligence, freezing and confiscation	Assets finally recovered, disrupted cash flow
Sector regulation	Internet, fuels, fintech, construction, health, procurement	Prices, entry, ownership transparency
Prison governance	Intelligence, lawful communications control, work and reintegration	Violence, illicit communication, recidivism

Source: MJSP, ENFOC and ENCCLA; SENAPPEN/CNCP, National Criminal and Penitentiary Policy Plan 2024–2027.

Discussion: what outcome would demonstrate greater state capacity?

Choose one instrument and the final outcome that should be used to evaluate it.

Operations

Activity measure

Arrests

Intermediate output

Recovered assets

Financial outcome

Market entry

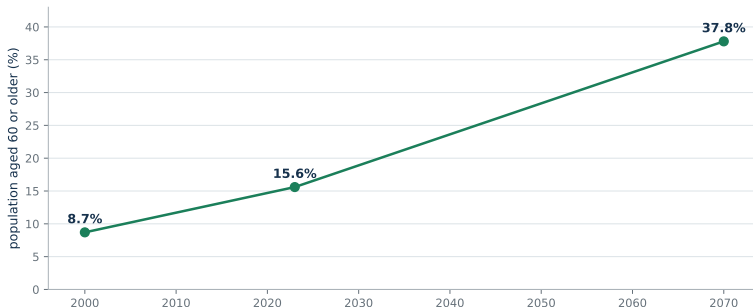
Development outcome

- ▶ Add one safeguard against abuse or displacement to another territory.
- ▶ Explain why your chosen outcome is closer to welfare than the activity count.

Micro scorecard: strong examples, weak diffusion

Area	Evidence of capability	Binding margin
Technology	Agricultural research and digital finance	Diffusion beyond frontier sectors
Firms	Globally competitive exporters and banks	Long tail of small, informal producers
Tax system	Reform establishes a modern VAT structure	Transition, exemptions, and implementation
Trade	Competitive commodity and selected manufacturing exports	Low integration and costly inputs
Workers	Broad enrollment and recent labor-market strength	Learning gaps and informality
Infrastructure	Scalable concessions and digital public rails	Logistics, sanitation, and local coordination
State capacity	Strong agencies and data systems in selected areas	Uneven local capacity and territorial coercion

Demography raises the return to productivity now



- ▶ The working-age share will no longer provide the same contribution to growth.
- ▶ Health, pensions, care, lifelong learning, and later-life employment become linked policies.

Source: IBGE, Population Projections 2024. The 2070 value is a projection, not an observed outcome.

Three clocks tighten the development constraint

Demography

Population peaks around 2041. The 60+ share rises from 15.6% in 2023 to 37.8% in 2070.

Climate

Droughts, floods, heat, and land-use pressure affect capital, food, energy, and cities.

Fiscal space

Aging raises health and pension pressure while adaptation and infrastructure require investment.

Future growth must rely more on productivity because labor-force expansion is ending.

Source: IBGE, Population Projections 2024; OECD, Economic Survey Brazil 2023.

Brazil has a green advantage and a territorial emissions problem

Energy advantage

86.6%

renewable domestic electricity supply in 2025

- ▶ Hydro, wind, solar, biomass, and biofuels create options for low-carbon production.

Emissions structure

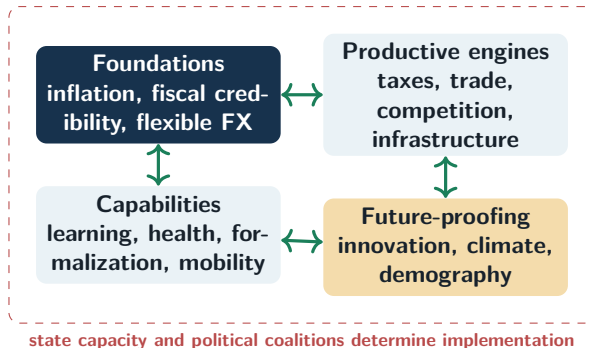
about 70%

from land use and agriculture in the 2022 inventory

- ▶ Enforcement, livestock productivity, restoration, and urban adaptation are central.

Source: EPE, BEN 2026; MCTI, National Emissions Inventory, 2022 results.

The agenda is a package of complements



Priority exercise: allocate a reform budget

Your group has R\$100 of political capital. Allocate it across the four pillars.

Foundations	Productive engines	Capabilities	Future-proofing
R\$_____	R\$_____	R\$_____	R\$_____

- ▶ Name one complement without which your largest allocation would fail.
- ▶ Name one group or region that bears the transition cost.

The diagnosis is complementarity, not a single missing reform

- ▶ Macroeconomic stability protects planning horizons and keeps exchange-rate shocks from becoming persistent inflation.
- ▶ A flexible exchange rate, credible fiscal policy, and external buffers improve shock absorption.
- ▶ Productivity requires firms to adopt better methods and resources to move toward productive uses.
- ▶ State capacity protects rights, regulates markets, delivers services, and confronts territorial coercion.
- ▶ Inclusion, demography, and climate affect both the returns to reform and the coalition that can sustain it.

Questions and discussion